



college of health sciences
school of medicine
Department of Anatomy

Introduction to Human Anatomy

Objectives:

- Define what is anatomy
- Describe history of anatomy
- Describe the divisions and approaches of studying anatomy
- Recognize anatomicomedical terminologies and apply them in studying anatomy
- Describe anatomical variations

Cont...

- ✓ Define what is histology
- ✓ Describe the cell structure
- ✓ Explain the tissue structure
- ✓ Describe the procedures of tissue preparation

Cont...

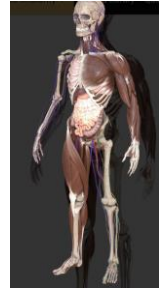
- Explain embryological terminologies
- Describe developmental periods
- ✓ Explain the main developmental changes in each period
- Understand critical periods of human development

Definition

- Branch of Biological sciences concerned with the study of the structure of organisms and their parts
- Derived from the Greek word **anatomē** meaning “to cut up” or taking apart
- The Latin word **dissection** has a similar meaning
- deals with the **structural** organization of living things
- it is the **structure** in which the **functions** of life occur
- It is the science of the structure of human body and the **relations** of its parts.

Cont...

- Anatomy has many sub-disciplines, and used in many different fields
 - In **general**, there are two main types of anatomy;
1. **Gross or macroscopic anatomy**



Cont...

2. Microscopic anatomy

- Cytology – study of the cell
- Histology – study of tissues



Cont...

Developmental Anatomy

- Is the study of **developmental** changes of the body before birth
- Provide scientific basis for understanding mechanisms underlying both **normal** and **abnormal** development
- It is also important to provide a logical basis for understanding the overall organization of the human body



Cont...

Uses

- Helps us understand human **anatomy** better
- Helps us understand why some children are born with structures that are **abnormal**
-
- help us prevent or **treat** such **abnormalities**

Approaches of studying Anatomy

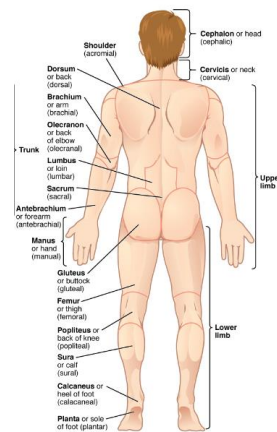
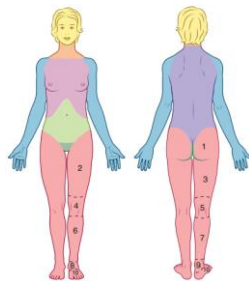
- three main approaches

I. Regional**II. Systemic****III. Clinical (applied)**

Cont....

I. Regional approach

- Also known as **topographical anatomy**
- method of studying the body's structure by focusing attention on a **specific part**
- Emphasize the **arrangement** and **relationships** of the various structures (muscles, nerves, arteries, etc.) on a **specific part**



Cont....

- Consider the organization of the human body as **major parts** consisting of:

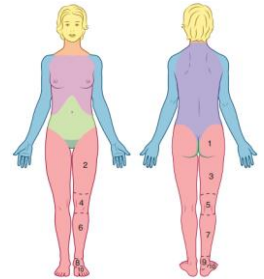
- ✓ Head
- ✓ Neck
- ✓ Trunk
- ✓ paired upper and lower limbs



B Posterior view

Cont....

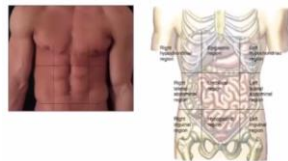
- Involves dissection
- All the major parts also further **subdivided** into regions and zones
- E.g. **trunk** subdivided into: thorax, abdomen, back, and pelvis/perineum),
- Lower limb** subdivided into: gluteal region, thigh region, knee region, leg region, ankle region, foot region



Cont....

➤ Surface anatomy

- also known as **superficial anatomy** and visual anatomy
- part of the study of **regional anatomy**
- It is the study of **internal organs** with respect to the surface of the body
- in clinical practice, to distinguish any **unusual** or **abnormal** findings



Cont....

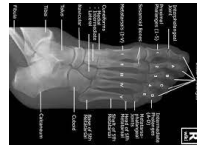
- Physical examination** is the clinical application of surface anatomy
- the evident and **palpable** surface features of the body are described
- understanding of the anatomy of the structures **beneath** the surface is required



Cont....

➤ Radiographic anatomy

- study of anatomy through the use of radiographic films
- Study of **deep** structures and **abnormalities** in a living person can be possible by means of radiographic and sectional imaging and endoscopy
- provides useful information about normal structures in **living** individuals



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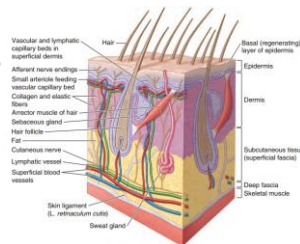
II. Systemic anatomy

- is the study of the body's organ systems that work together to carry out complex functions.

Cont....

➤ Integumentary system

- dermatology
- consists of the **skin** and its **appendages** and the subcutaneous tissue just beneath it



Cont....

➤ Skeletal system

- Osteology
- consists of **bones** and **cartilage**

➤ Articular system

- Arthrology
- consists of **joints** and their associated **ligaments**



Cont....

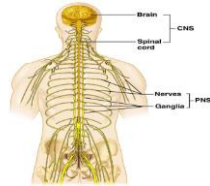
➤ Muscular system

- Myology
- consists of:
 - skeletal muscle
 - smooth muscle
 - cardiac muscle



➤ Nervous system

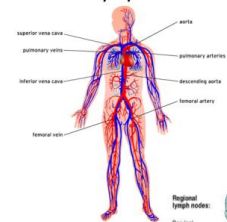
- Neurology
- consists of the **central** and the **peripheral** nervous system



Cont....

➤ Circulatory system

- **Angiology**
- consists of the **cardiovascular** and **lymphatic** systems

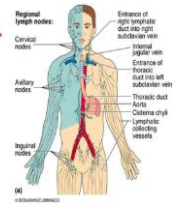


✓ Cardiovascular system

- **Cardiology**
- consists of the **heart, blood** and **blood vessels**

✓ lymphatic system

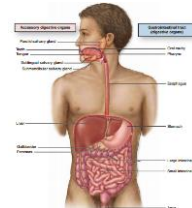
- is a network of lymphatic vessels



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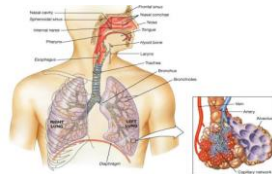
➤ alimentary or digestive system

- Gastroenterology
- consists of the digestive tract from the **mouth** to the **anus**, with all its associated organs and **glands**



➤ Respiratory system

- Pulmonology
- consists of the **air passages** and **lungs**



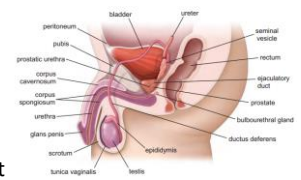
Cont....

➤ Urinary system

- **urology**
- consists of the **kidneys, ureters, urinary bladder**, and **urethra**

➤ Genital (reproductive) system

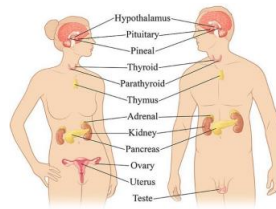
- **gynecology** for females;
- **andrology** for males
- consists of the **gonads** (ovaries and testes), the **ducts**, and the **genitalia** that enable their union



Cont....

➤ Endocrine system

- Endocrinology
- consists of specialized ductless structures that secrete hormones



Cont...

III. Clinical anatomy (applied anatomy)

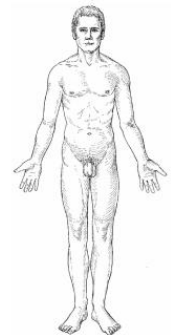
- It incorporates the **regional** and **systemic** approaches to studying anatomy and stresses clinical application
- involves **inverting** or **reversing** the thought process typically followed when studying regional or systemic anatomy
- For example,
 - ✓ instead of thinking, "The action of this muscle is to . . .," clinical anatomy asks, "How would the **absence** of this muscle's activity be manifest?"
 - ✓ Instead of noting, "The . . . nerve provides innervation to this area of skin," clinical anatomy asks, "**Numbness** in this area indicates a lesion of which nerve?"

Anatomicomedical terminologies

- International **anatomical** terminology
- enables **precise** communication among healthcare professionals and scientists worldwide
- describe anatomical locations, preventing both **confusion** and **misunderstanding**
- ✓ To describe their complaints health professionals must also know the common and **colloquial** terms

Anatomical Position

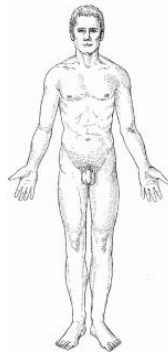
- **consistent** position
- ensuring that descriptions are not **ambiguous**
- One must visualize this position in the **mind** when describing patients (or cadavers)
- adopted **globally** for anatomicomedical descriptions



Cont...

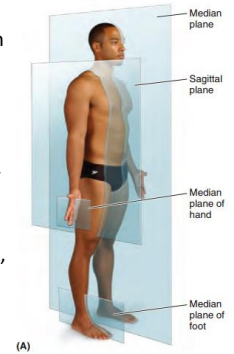
Cont....

- In anatomical position:
- the person **standupright** with the:
- ✓ head, eyes, and toes directed anteriorly (**forward**),
- ✓ arms adjacent to the sides with the palms facing **anteriorly**, and
- ✓ lower limbs close together with the feet **parallel**



Anatomical Planes

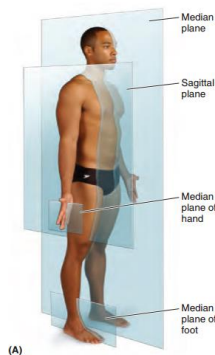
- **Imaginary** planes that intersect the body in the anatomical position
- Anatomical descriptions are based on these planes
- Four imaginary planes:
- 1. **Median (median sagittal) plane**
 - **vertical** plane
 - pass **longitudinally** through the body
 - divides the body into **right** and **left** halves
 - defines the **midline** of the head, neck, and trunk where it intersects the surface of the body



Cont....

2. Sagittal planes

- **Vertical** planes
- passing through the body **parallel** to the median plane
- **paramedian** plane is a plane parallel and near to the median plane



Cont....

3. Frontal (coronal) planes

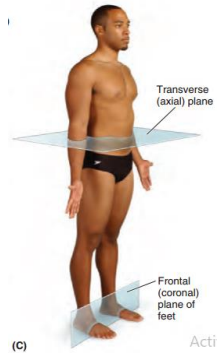
- **Vertical** planes
- passing through the body at **right angles** to the median plane
- Divide the body into **anterior** and **posterior** parts



Cont....

4. Transverse planes

- Horizontal planes
- passing through the body at right angles to the median and frontal planes
- Divide the body into superior and inferior parts



- The main use of anatomical planes is to describe sections

➤ Longitudinal sections

- run lengthwise or parallel to the long axis of the body or of any of its parts



Cont....

Cont....

➤ Transverse sections, or cross sections

- are slices of the body or its parts that are cut at right angles to the longitudinal axis of the body or of any of its parts

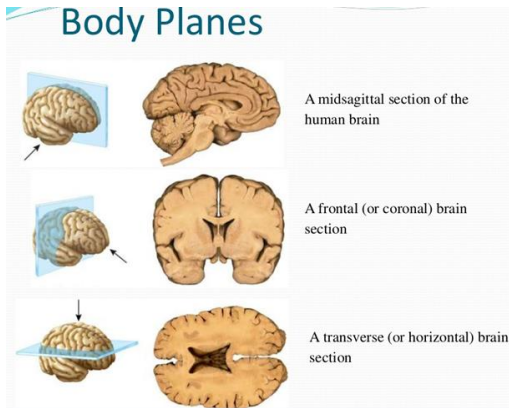


Cont....

➤ Oblique sections

- are slices of the body or any of its parts that are not cut along the previously listed anatomical planes





Terms of Relationship and Comparison

- Describe the **relationship** of **parts** of the body
- **Compare** the position of two structures relative to each other

Cont....

Superior

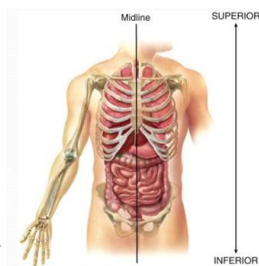
- refers to a structure that is nearer the **vertex** or toward the **head**

- **Above**

Inferior

- refers to a structure that is nearer the sole of the **foot**
- **Away** from the head or **toward** the lower part of a structure or the body

- **Below**



Cont....

✓ Anterior (ventral)

- denotes the **front** surface of the body

✓ Posterior (dorsal)

- denotes the **back** surface of the body or nearer to the back



Cont....

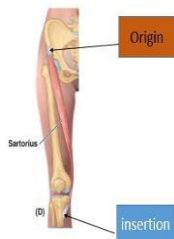
- **External** means **outside** of or **farther** from the center of an organ or cavity
- **Internal** means inside or **closer** to the **center**, independent of direction
- **Combined** terms describe intermediate positional arrangements:
 - **Inferomedial** means nearer to the **feet** and **median** plane
 - **Superolateral** means nearer to the **head** and **farther** from the median plane

Cont...

- **Medial** indicates that a structure is **nearer** to the **median plane** of the body
- **Lateral** stipulates that a structure is farther **away** from the **median plane**



- ✓ **proximal** closer to origin of a structure
- ✓ **distal** further away from origin of a structure



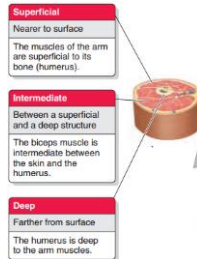
Cont...

- **Dorsum** usually refers to the **superior** or **posterior** (back) surface of any part that protrudes anteriorly from the body
 - It is also used to describe the back of the **hand**
- **The palm** refers to the **flat** of the hand, exclusive of the **thumb** and other **fingers**, and is the opposite of the dorsum of the hand



Cont...

- **Superficial, intermediate, and deep** describe the position of structures relative to:
 - the **surface** of the body
 - The relationship of one structure to another underlying or overlying structure



Cont...

❖ **Terms of Laterality:**

- ✓ **Ipsilateral** something occurring on the **same** side of the body relative to another structure

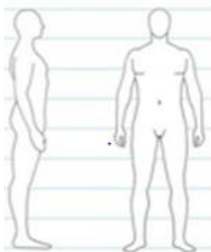


- ✓ **Contralateral** means occurring on the **opposite** side of the body relative to another structure



Cont...

- ✓ **Bilateral** means paired structures having right and left members
- ✓ **Unilateral** means occurring on one side only

**Terms in relation with the embryo****Ventral**

- **Anterior** surface of the embryo

Dorsal

- **Posterior** surface of the embryo
- cranial/rostral**

- Means toward the **head**

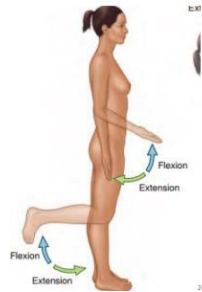
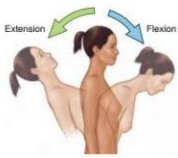
Caudal

- Means toward the **tail**

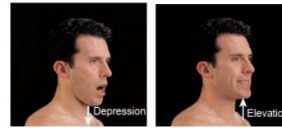
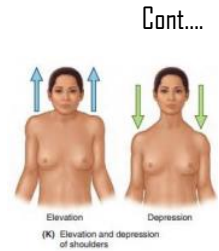


Terms of movement

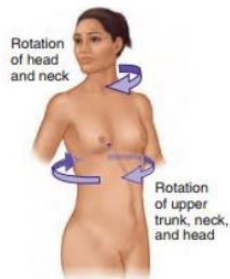
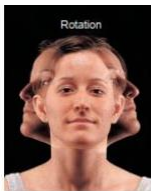
- ✓ **Flexion** indicates **bending** or **decreasing** the angle between the bones or parts of the body
- ✓ **Extension** indicates **straightening** or **increasing** the angle between the bones or parts of the body



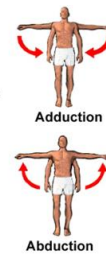
- ✓ **Elevation** raises or moves a part **superiorly**
- ✓ **Depression** lowers or moves a part **inferiorly**



- ✓ **Rotation** involves **turning** or **revolving** part of the body around its longitudinal axis



- ✓ **Abduction** means moving **away** from the **median** plane except for the digits
- ✓ **Adduction** means moving **toward** it **median** plane



Cont...

- ✓ **abduction of the digits (fingers or toes)** means **spreading** them apart
- moving the other fingers away from the neutrally positioned **3rd** (middle) finger
- or moving the other toes away from the neutrally positioned **2nd** toe
- ✓ **Adduction** of the digits is the **opposite**



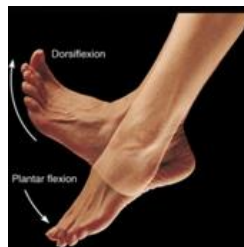
Cont...

- ✓ **Eversion** refers to movement of the **sole** of the foot **away** from the median plane
- turning the sole **laterally**
- ✓ **Inversion** describes movement of the sole of the foot **toward** the median plane



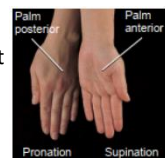
Cont....

- **Dorsiflexion**
- Ankle movement bringing the foot to the shin
- **Plantarflexion**
- Ankle movement pointing downwards



Cont...

- **Pronation**
- is rotation of the radius **medially**
- the **palm** of the hand faces **posteriorly**
- Its **dorsum** faces **anteriorly**
- **Supination**
- is the opposite rotational movement



Cont...

- **Opposition**

- is the movement by which the pad of the 1st digit (thumb) is brought to another digit pad

- **Reposition**

- describes the opposite movement



Cont....

- ✓ **Circumduction**

- combined movements of flexion, extension, abduction, adduction medial and lateral rotation circumscribe a cone



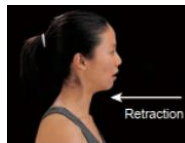
Cont...

- ✓ **Protrusion**

- describes a movement **forwards**
- protruding the mandible, lips, or tongue

- ✓ **Retrusion**

- describes a movement **backward**
- retruding the mandible, lips, or tongue



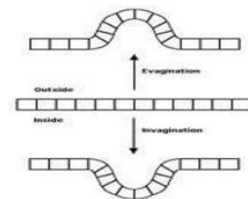
Cont...

- Invagination**

- Is projection inside

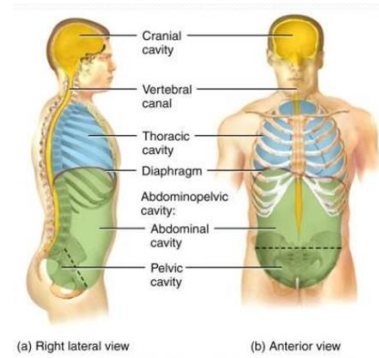
- Evagination**

- Is projection out side



Body Cavities

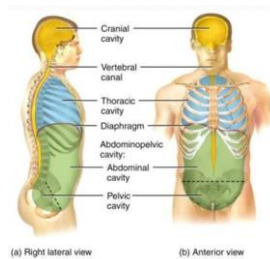
- Spaces within the body which contain **vital** organs
- Help **protect**, separate, and support internal organs
- Permit **changes** in size and shape of internal organs
- Two sets of internal body cavities
 - **Dorsal** cavity
 - **Ventral** cavity



Cont...

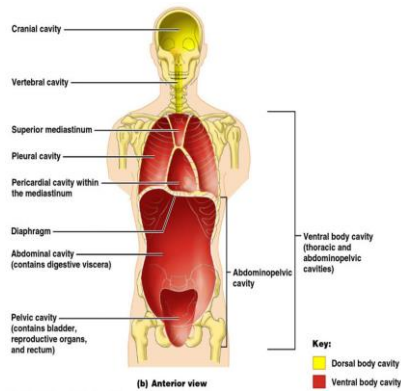
➤ Ventral cavity

- Houses the **internal** organs (viscera)
- divided into two subdivisions:
 - Thoracic
 - Abdominopelvic cavities



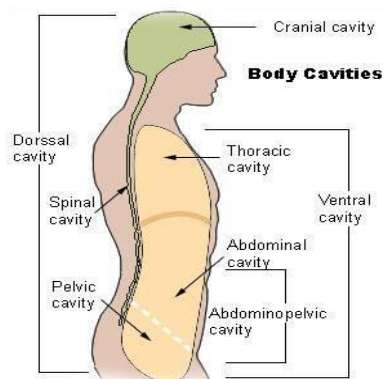
Thoracic cavity (chest cavity)

- The upper ventral cavity
- contains the **heart**, **lungs**, **trachea**, **esophagus**, large blood vessels, and nerves
- bound laterally by the **ribs** (covered by costal pleura) and the diaphragm **caudally** (covered by diaphragmatic pleura)
- Divided into 2 **pleural** and 1 **pericardial** cavities by mediastinum

6
5

Abdominopelvic cavity

- separated from the thoracic cavity by the dome-shaped **diaphragm**
- It is composed of two subdivisions:
 - I. Abdominal cavity**
 - Contains most of the **GIT** as well as the **kidneys** and **adrenal glands**
 - Bound cranially by the diaphragm, laterally by the body wall, and caudally by the pelvic cavity
 - II. Pelvic cavity**
 - lies within the pelvis
 - contains the bladder, reproductive organs, and rectum



Dorsal cavity

- protects the **nervous system**
- smaller of the two main cavities
- divided into two subdivisions:
 - **Cranial cavity**
 - within the skull
 - encases the brain
 - **Vertebral cavity**
 - runs within the vertebral column
 - encases the spinal cord

Body membranes

- are thin **sheets** of tissue that cover or line the **body cavities**, and cover organs within the cavities in hollow organs
- categorized into **epithelial** and **connective tissue membranes**

I. Epithelial Membranes

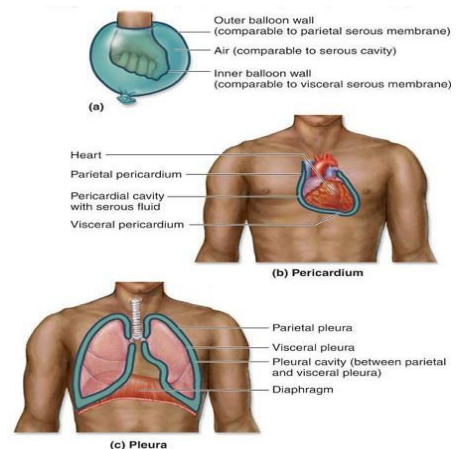
- consist of **epithelial** tissue and the connective tissue to which it is attached
- The two main types of epithelial membranes are:
 - ✓ mucous membranes
 - ✓ and serous membranes

I Mucous Membranes

- line the body cavities that **open to the outside**
- Line the entire **digestive tract**, **respiratory**, and **reproductive tracts** and much of the **urinary** system
- The epithelial layer of a mucous membrane secretes **mucus**
- mucus prevents the cavities from **drying out**
- It also **traps** particles in the respiratory passageways, **lubricates** and **absorbs** food as it moves through the GIT, and secretes **digestive** enzymes

II. Serous Membranes

- line body cavities that **do not open directly to the outside**
- the organs located in those cavities covered by a thin layer of **serous fluid** that is secreted by the epithelium
- It **lubricates** the membrane and **reduces** friction and abrasion when organs in the cavity move against each other or the cavity wall
- Serous membranes have special names given according to their location;
 - ✓ E.g. the serous membrane that lines the thoracic cavity and covers the lungs is called **pleura**



2. Connective Tissue Membranes

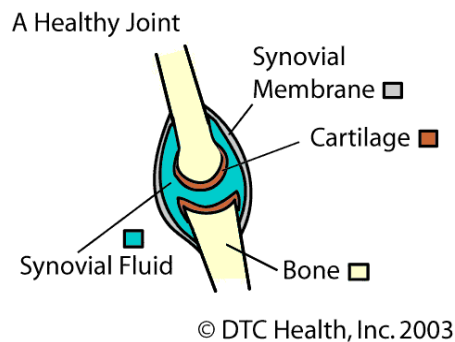
- Contain only **connective** tissue
- **Synovial membranes** and **meninges** belong to this category

Meninges:

- the connective tissue covering on the **brain** and **spinal cord**
- They provide **protection** for these vital structures

Synovial Membranes

- line the cavities of **freely movable joints** such as the *shoulder, elbow, and knee*
- Like serous membranes, they line cavities that do **not open** to the outside
- Unlike serous membranes, they do not have a layer of epithelium
- Synovial membranes secrete **synovial fluid** into the joint cavity, and this **lubricates** the **cartilage** on the ends of the bones so that they can move freely without friction



Levels of Structural Organization of Human Body

✓ Chemical level

- Includes all atoms and molecules essential for maintaining life

- Major atoms:- **C, H, O, N, Ca, K & Na**

- Molecules:- **proteins, carbohydrates, fats & vitamins**

✓ Cellular level

- Cells are the basic structural & functional unit of life
- Cells have 4 principal parts

Cont....

1. Plasma (cell) membrane

- outer limiting membrane separating the cell's internal components from the extracellular materials

2. Cytosol

- Thick semifluid portion of the cytoplasm
- Cytoplasm refers to all cellular contents b/n the plasma membrane & the nucleus
- Contains many soluble proteins & enzymes, nutrients, ions and other small molecules
- Organelles and inclusions are suspended in it

Cont....

3. Organelles

- Highly organized structures with characteristic shapes that are specialized for specific cellular activities
- E.g. mitochondria, GA, ER, lysosomes, etc.

4. Inclusions

- Temporary structures that contain secretions & storage products of the cell
- Such as lipofuscin, hemosiderin, hemoglobin, glycogen, lipid droplets, etc

Cont...

✓Tissue level

- Tissues are groups of similar cells and substance surrounding them that work together to perform a particular function

4 basic types of tissues:

- Epithelial tissue
- Muscle tissue
- Connective tissue
- Nervous tissue

Cont....

✓Organ level

- Organs are composed of two or more d/t tissues that have specific functions & usually have recognizable shape

✓System level

- A system consists of several related organs have a common function E.g. Digestive system:- breakdown of food & absorption

✓Organism level

- All parts of the body functioning with one another comprise the total organism

Anatomical variation

- Normal flexibility in the **topography** and **morphology** of body structures
- Anatomy books describe the structure of the body as it is usually observed in people—that is, the **most** common pattern.
- However, occasionally a particular structure demonstrates so much **variation** within the normal range

✓ Physical, sexual, racial, genetic, etc.

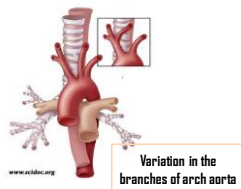
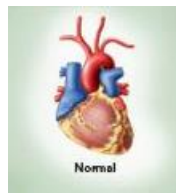
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Cont...

- Individual variation must be considered in physical **examination, diagnosis, and treatment**
- **Surgeons** should have to know **clinically** significant variations
- Clinical awareness of anatomical variations is also important proper interpretation of **imaging modalities**, pain management therapies and others

Cont....

- variation exists in the:
 - ✓ patterns of branching of veins, arteries, and nerves
 - *Veins vary the most and nerves the least*
 - ✓ size, shape, and form of the attachments of **muscles**
- bones of the skeleton vary not only in their basic shape but also in lesser details of surface structure



Thank you



Thank you